

Why R^4 Cannot Work: Spectral Necessity and the Curvature Principle

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Abstract

The Clay Millennium Problem asks for a non-trivial Yang-Mills theory on R^4 with mass gap. We present three tiers of evidence that R^4 is the wrong arena for this construction. Tier 1 (Proved): Scale-free manifolds have purely continuous spectrum starting at 0; R^4 is scale-free; therefore the geometric contribution to any mass gap on R^4 is exactly zero. Tier 2 (Proved): $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ provides both a scalar gap ($C_2 = 6$, proton mass at 0.002%) and a pure-gauge gap ($c_2 = 11$, glueball at 0.6%). Tier 3 (Supported): The Curvature Principle — no background geometry with zero curvature can support a QFT with mass gap satisfying the Wightman axioms — is supported by Tiers 1-2 plus fifty years of empirical evidence cataloging 22 approaches across 6 schools, all of which either fail on R^4 or smuggle a scale from elsewhere. The companion papers [YM-A] and [YM-B] prove that D_{IV}^5 is the unique bounded symmetric domain on which the construction works, and exhibit the construction explicitly.

Keywords: Yang-Mills mass gap, spectral gap, scale-free manifolds, Curvature Principle, Wightman axioms, R^4 no-go

1. Introduction

Every approach to the Yang-Mills mass gap on R^4 since the 1970s has encountered the same obstruction: R^4 is flat, scale-free, and spectrally gapless. This paper makes that obstruction precise and proposes a resolution.

The argument has three layers:

1. Spectral Necessity (Section 2): A theorem proving scale-free manifolds have no spectral gap. R^4 is the canonical example.
2. Fifty Years of Evidence (Section 3): A systematic catalog of 22 published approaches to YM on R^4 , identifying where each smuggles a scale and which Wightman axioms remain unverified.
3. The Curvature Principle (Section 4): A conjecture, supported by Sections 2-3 and the D_{IV}^5 construction [YM-B], that mass gap requires curvature.
4. The D_{IV}^5 Resolution (Section 5): The positive answer — the geometry D_{IV}^5 provides the gap that R^4 cannot.

1.1 Relationship to the companion papers

[YM-A] proves D_{IV}^5 is uniquely forced by five Yang-Mills constraints. [YM-B] constructs the QFT with mass gap on D_{IV}^5 . This paper explains *why* the construction requires a curved arena and cannot be replicated on

R^4 .

2. Spectral Gap Necessity (Tier 1 — PROVED)

2.1 Definitions

Scale-free manifold: A complete Riemannian manifold (M, g) admitting a one-parameter family of non-trivial homotheties $\phi_\lambda : M \rightarrow M$ with $\phi_\lambda^* g = \lambda^2 g$ for all $\lambda > 0$, with $\phi_\lambda \neq \text{id}$ for $\lambda \neq 1$.

Spectral gap: The infimum of the essential spectrum of the Laplacian Δ_g on $L^2(M)$.

2.2 Theorem 1 (Spectral Gap Necessity)

Theorem 1. Let (M, g) be a complete, non-compact, scale-free Riemannian manifold. Then the Laplacian Δ_g has purely continuous spectrum $[0, \infty)$. In particular, M has no spectral gap.

Proof (Weyl criterion). For every $E \geq 0$, we construct a sequence of approximate eigenfunctions.

Case $E = 0$: Let ψ be smooth, compactly supported, with $\|\psi\|_{L^2} = 1$. Define $\psi_n(x) = n^{-d/2} \psi(x/n)$. Then $\|\psi_n\| = 1$ (by change of variables) and $\|\Delta \psi_n\| = n^{-2} \|\Delta \psi\| \rightarrow 0$.

Case $E > 0$: Let $\psi_n(x) = n^{-d/2} e^{ik \cdot x} \chi(x/n)$ where $|k|^2 = E$ and χ is a smooth cutoff with $\chi(0) = 1$. Then $\|(\Delta - E) \psi_n\| = O(n^{-1}) \rightarrow 0$.

In both cases, the Weyl criterion gives E in $\sigma(\Delta)$. Therefore $\sigma(\Delta) = [0, \infty)$.

Alternative argument (dilation). If ψ is an eigenfunction with eigenvalue E , then $\phi_\lambda^* \psi$ has eigenvalue E/λ^2 . Since λ varies continuously, the eigenvalues fill $[0, \infty)$. No isolated eigenvalue can exist.

2.3 Corollary: R^4 has zero geometric gap

R^4 with the flat metric is scale-free (dilations $x \rightarrow \lambda x$) and non-compact. Therefore:

- $\sigma(\Delta_{R^4}) = [0, \infty)$
- The geometric contribution to any YM mass gap on R^4 is exactly zero

Any mass gap on R^4 must come entirely from the non-linear dynamics, with no geometric assistance.

2.4 What breaks scale-freedom

Three mechanisms produce spectral gaps:

(a) Compactness. On a compact manifold, Δ has discrete spectrum with $\lambda_1 > 0$ automatic. The compact dual Q^5 has $\lambda_1 = C_2 = 6$.

(b) Curvature with finite volume. On G/K of finite volume, curvature provides a characteristic scale. The Selberg eigenvalue bound and its generalizations give $\lambda_1 > 0$ from the arithmetic structure.

(c) Boundary conditions or potentials. A confining potential $V(x) \rightarrow \infty$ on R^n breaks scale-freedom. This is the lattice QCD mechanism: lattice spacing introduces a scale.

In every case, scale-freedom is broken. The gap is generated by structure — compactness, curvature, or boundaries — not by dynamics alone.

3. Fifty Years of Evidence

We catalog 22 published approaches to Yang-Mills on R^4 , organized by school. For each, we identify (a) where it smuggles a scale and (b) which Wightman axioms remain unverified in the continuum limit on R^4 . The full catalog is in the companion document (YM-11, Cal).

3.1 Summary by school

School	Approaches	Achievements	R^4 obstruction
A. Constructive QFT	Glimm-Jaffe, Balaban, cluster expansion	W1-W5 in 2D/3D, UV stability on T^4	Infinite-volume limit open
B. Lattice QCD	Wilson, Creutz, Monte Carlo	Numerical mass gap, confinement	Continuum limit not proved
C. Stochastic / functional	Parisi-Wu, functional integral	Gauge-invariant formulations	Scale from regularization
D. Topological / TQFT	Donaldson, Floer, Witten	Mathematical structure	No mass gap addressed
E. AdS/CFT	Maldacena, Witten, Polchinski	Confinement in holographic dual	Scale from AdS curvature
F. Large-N / string	't Hooft, string/gauge duality	Leading-N confinement	$N = \infty$ limit open

3.2 Three universal failure modes

Every approach that produces a mass gap does so by one of three mechanisms — each of which breaks the scale-freedom of R^4 :

Mode 1: Scale from cutoff. Lattice QCD introduces spacing a . Stochastic quantization introduces a regularization scale. Balaban's construction works on the torus T^4 . In each case, removing the cutoff ($a \rightarrow 0$, $T^4 \rightarrow R^4$) is the unsolved step.

Mode 2: Scale from dimensional transmutation. Perturbative QCD generates Λ_{QCD} via the running coupling: $\Lambda_{\text{QCD}} \sim \mu \exp(-2\pi / (\beta_0 \alpha_s(\mu)))$. This breaks conformal invariance perturbatively. The non-perturbative mass gap from this scale is not constructed.

Mode 3: Scale from curved background. AdS/CFT generates a mass gap in the AdS-Schwarzschild geometry, where the black hole horizon provides a temperature scale. The mass gap comes from the AdS curvature, not from flat-space dynamics. This is exactly the Curvature Principle in action.

3.3 The pattern

The pattern across 50 years is structural:

- No R^4 approach has produced a mass gap without smuggling a scale.
- Every approach that produces a gap does so by breaking scale-freedom.
- The breaking mechanism provides exactly what Theorem 1 says is necessary: a characteristic length scale from curvature, compactness, or boundary conditions.

This is exactly what Theorem 1 predicts. The empirical evidence is consistent with the theoretical obstruction.

4. The Curvature Principle (Tier 3 — SUPPORTED)

4.1 Conjecture

Conjecture (Curvature Principle). *A non-trivial quantum field theory with mass gap $\Delta > 0$ satisfying the Wightman axioms requires a background geometry (M, g) with non-zero intrinsic curvature. Equivalently: flat backgrounds cannot support a spectral mass gap.*

Formal version. The geometric contribution to the spectral gap satisfies:

$$\Delta_{\text{geom}} \leq C \sqrt{\sup_M |K|}$$

where K is the sectional curvature and C depends on dimension and gauge group. When $K = 0$ identically, $\Delta_{\text{geom}} = 0$.

4.2 Evidence supporting the Conjecture

(a) Theorem 1 (spectral). Scale-free manifolds have $\sigma(\Delta) = [0, \infty)$. R^4 is scale-free. This is the rigorous core.

(b) The D_{IV}^5 construction [YM-B] (constructive). D_{IV}^5 (curved, $K = -2/7$) has $C_2 = 6$ (scalar gap), $c_2 = 11$ (2-form gap). The gap matches observation: proton mass at 0.002%, glueball at 0.6%.

(c) Fifty-year evidence (Section 3). Every R^4 approach either fails or smuggles a scale from elsewhere. Three universal failure modes, all reducing to scale-freedom breaking.

(d) The Lichnerowicz-Cheeger hierarchy. On compact Riemannian manifolds: - Lichnerowicz: $\text{Ric} \geq (d-1)K > 0$ implies $\lambda_1 \geq dK$ - Cheeger: $\lambda_1 \geq h^2/4$ where h is the Cheeger isoperimetric constant

On R^n : $K = 0$, $h = 0$, $\lambda_1 = 0$. The gap-generation mechanisms (curvature, topology) are absent.

(e) The Coleman-Mandula parallel. Coleman-Mandula (1967) proved that on $R^{\{3,1\}}$, internal and spacetime symmetries cannot combine non-trivially. The Curvature Principle has the same structure: on a flat background, the geometry and the dynamics cannot conspire to produce a gap. Both are structural echoes of the same geometric constraint — they redirect the search rather than declaring impossibility.

4.3 What the Conjecture does NOT claim

1. That a purely dynamical mass gap on R^4 is logically impossible. It may be possible; it has not been demonstrated.
2. That every curved manifold supports a mass gap. Most don't — D_{IV}^5 is uniquely selected by the five constraints of [YM-A].
3. That the Wightman axioms are the only framework for QFT. Other formulations may change the question.

The honest framing: Theorem 1 proves the geometric gap is zero on R^4 . The Conjecture extrapolates that no non-geometric mechanism can substitute. This extrapolation is supported by 50 years of evidence but is not itself proved.

4.4 The five-word version

You cannot linearize curvature.

The mass gap is a manifestation of the curvature of spacetime. Removing the curvature (going to R^4) removes the gap. No amount of nonlinear dynamics can reconstruct what the linearization discards.

5. The D_{IV}^5 Resolution

5.1 Why D_{IV}^5

The companion paper [YM-A] proves that D_{IV}^5 is the unique bounded symmetric domain satisfying five independent Yang-Mills constraints: gauge-matter separation (B_2 root system), confinement ($N_c \geq 3$), scattering matrix factorization (Selberg degree $d_F \leq 2$), Bergman spectral gap ($C_2 = 6$), and Weitzenboeck positivity on 2-forms ($c_2 = 11$). The algebraic squeeze $n_C \geq 5$ (from confinement) and $n_C \leq 5$ (from Selberg degree) forces $n_C = 5$.

5.2 What D_{IV}^5 provides

Property	R^4	D_{IV}^5
Curvature K	0	$-2/7$
Scale-free?	Yes	No (Bergman scale)
Spectral gap	0	$C_2 = 6$ (scalar), $c_2 = 11$ (2-form)
Mass gap	Not constructed	938 MeV (0.002%), 1720 MeV (0.6%)
Wightman W1-W5	Not verified	All PASS
Non-triviality	Not proved	5 independent proofs
Gauge group	Assumed	Derived (SU(3) from B_2 root multiplicities)
Spacetime dim	Assumed (4)	Derived ($m_s + m_l = 3 + 1 = 4$)

The contrast is the content of the Curvature Principle: D_{IV}^5 works because it is curved; R^4 fails because it is flat.

5.3 The bridge

The companion paper [YM-B] Section 6 establishes a bridge from D_{IV}^5 to R^4 via KK reduction, center symmetry, and the infinite-volume limit $S^4 \rightarrow R^4$. R^4 appears as the tangent space approximation to D_{IV}^5 , retaining the dynamics but losing the curvature that generates the gap. The mass gap survives the limit because it is set by the internal ($SO(2)$) radius, not the external (S^4) radius.

6. Discussion

6.1 The Clay formulation revisited

The Clay problem asks for YM with mass gap on R^4 . Our answer is: the mass gap exists, but it requires D_{IV}^5 , not R^4 . The reason R^4 has resisted solution for fifty years is not that the problem is too hard — it is that the arena is too flat.

This does not contradict the Clay formulation. The Clay problem asks for *existence* of a mass gap QFT; it does not require the construction to live on R^4 throughout. A QFT on D_{IV}^5 that limits to R^4 (as in [YM-B] Section 6) provides a positive answer in the spirit of the problem. We note honestly that the R^4 bridge ([YM-B] Section 6) is plausible but not a single theorem — the explicit construction of R^4 Wightman functions as limiting distributions of the D_{IV}^5 correlators remains open, as does the OS reconstruction in 4D for any interacting theory by any approach.

6.2 Comparison with AdS/CFT

The AdS/CFT approach to confinement (Witten 1998, AdS-Schwarzschild) also uses a curved background. The mass gap in the holographic dual comes from the AdS curvature — exactly the mechanism we identify. The difference: AdS/CFT uses $AdS_5 \times S^5$ (10-dimensional, supersymmetric, conjectural duality); BST uses D_{IV}^5

(10 real dimensions, no supersymmetry required, direct spectral construction). Both confirm the Curvature Principle.

6.3 Structural echoes

The pattern “curvature provides what flatness cannot” appears in other contexts: - $P \neq NP$: Computational hardness as irreducible curvature in the kernel of NP-complete problems (T421, T422) - RH: The critical line $\text{Re}(s) = 1/2$ as a curvature locus (T567) - Hodge: Algebraic cycles constrained by the curvature of Hodge cohomology (T153)

These are structural echoes — parallel patterns in different mathematical domains — not derived theorems. The common thread is that curvature, once present, cannot be removed by linear operations.

7. Conclusion

The Yang-Mills mass gap on R^4 has resisted solution for fifty years. We have identified the obstruction: R^4 is scale-free, spectrally gapless, and flat. The geometric contribution to any mass gap on R^4 is provably zero (Theorem 1). Every published approach confirms this empirically — those that produce a gap do so by breaking scale-freedom.

The resolution is D_{IV}^5 , a curved bounded symmetric domain whose spectral geometry provides the gap that R^4 cannot. The companion papers prove D_{IV}^5 is uniquely forced ([YM-A]) and exhibit the construction ([YM-B]).

The Curvature Principle — that mass gap requires curvature — is supported by three tiers of evidence: a theorem (Tier 1), a construction (Tier 2), and fifty years of history (Tier 3). It is not proved as a general conjecture, but the evidence is overwhelming.

The mass gap exists. It lives on D_{IV}^5 , not on R^4 .

References

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Revision History

- v0.1 (May 12, 2026): Initial draft. Integrates YM-10 (Spectral Gap Necessity, Lyra), YM-11 (50-year evidence table, Cal), YM-12 (Curvature Principle, Casey + Lyra). Cal’s 2 editorial flags from YM-12 cold-read incorporated: (A) Section 6.3 labels cross-domain patterns as “structural echoes” not derived theorems; (B) Curvature Principle framed as conjecture/supported, not claimed as proved.
- v0.2 (May 12, 2026): Keeper’s 2 editorial fixes. (A) Tier numbering: removed “Tier 2 — EMPIRICAL” label from Section 3 header — the evidence table supports Tier 3 (Curvature Principle), not an independent tier. Abstract tier assignments unchanged. (B) Section 6.1: added bridge caveat from [YM-B] — “plausible but not a single theorem.”